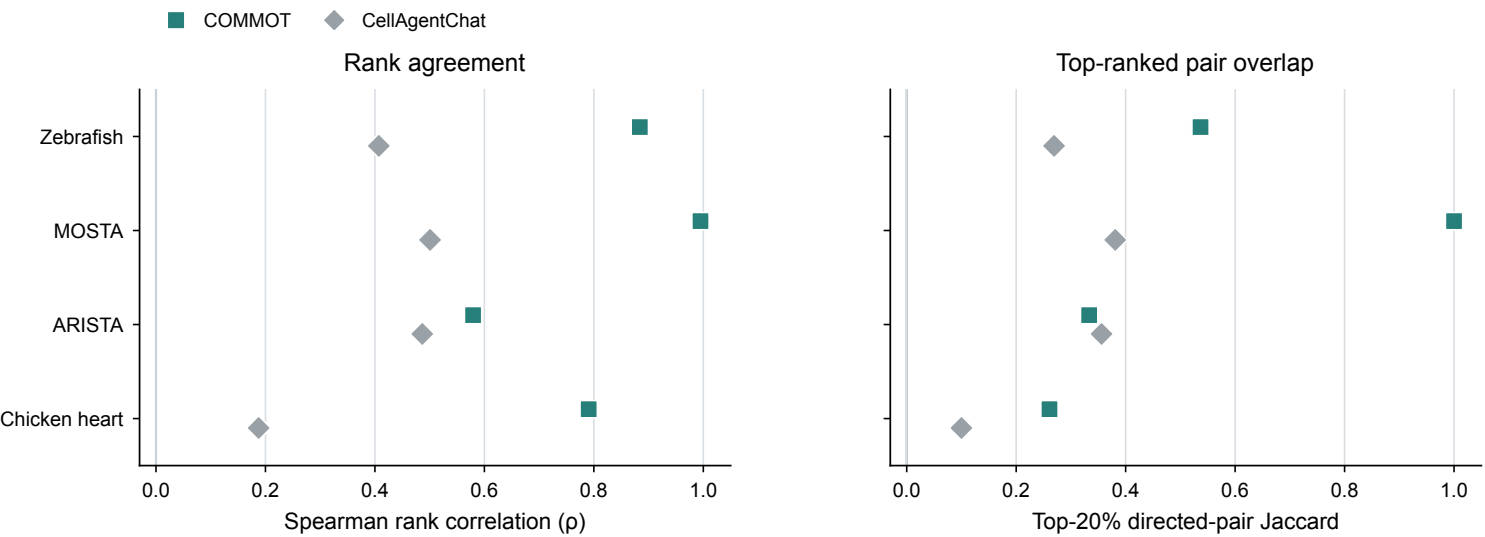


a Directed-pair consistency



b CytoBridge-selected LR axes

Dataset	CytoBridge selection				COMMOT comparison	
	Sender	Receiver	Selected LR axis	Database pathway	Same-pair LR rank	Pair percentile
Zebrafish	Pronephros	Musculature / YSL	COL1A2–SDC4	COLLAGEN	7 / 1,991	75.9
MOSTA	Muscle	Connective tissue	COL6A3–CD44	COLLAGEN	44 / 1,757	93.3
ARISTA	sfrpEGC	tinBL	PSAP–GPR37L1	PSAP	2 / 480	94.2
Chicken heart	Fibroblast cells	Valve cells	CD99–CD99	CD99	2 / 1,105	96.5

c NicheNet receiver targets for CytoBridge-selected axes

Dataset	CytoBridge selection		COMMOT and NicheNet results	
	Directed pair (sender; receiver)	LR axis Database pathway	COMMOT same-axis percentile	NicheNet-predicted receiver targets
Zebrafish	Sender: Spinal Cord Anterior Region Receiver: Nervous System	CDH2–CDH2 CDH	99.2	Ccnd1, Cbx5, Cdk1
MOSTA	Sender: Muscle Receiver: Connective tissue	COL6A3–CD44 COLLAGEN	99.3	Cd44
ARISTA	Sender: sstIN Receiver: mpEX	L1CAM–L1CAM L1CAM	99.1	EGR1, DDIT4, JUNB
Chicken heart	Sender: Valve cells Receiver: Fibroblast cells	LAMB1–DAG1 LAMININ	99.9	ID1, FN1, HNRNPR